

ACADEMIC EMPLOYMENT

New York University Stern School of Business

July 2025 – July 2026

Postdoctoral Researcher working on post-training for LLM's.

Columbia University Statistics Department

Jan. 2025 – June 2025

Postdoctoral Research Scientist working on efficient online outlier detection.

EDUCATION

Columbia University

2018–2024

PhD in Statistics

Stony Brook University

2014–2018

B.S. in Mathematics, magna cum laude

RESEARCH BACKGROUND

My research interests are reinforcement learning theory and post-training for LLM's. As a postdoc, currently I am working on proving a rigorous mathematical foundation for various empirical phenomena observed in real-life LLM post-training, such as step-size scaling and entropy collapse. Prior to this, during my PhD, I focused on statistical learning theory, with an emphasis on sequential-decision making (e.g., multi-armed bandits, online learning, RL) under dynamic changing environments. My works in this regard proved some of the current state-of-the-art regret guarantees in multiple settings such as multi-armed bandits, contextual bandits, and dueling bandits. During undergrad, I did pure math research in combinatorics and representation theory.

PREPRINTS AND PUBLICATIONS

(* denotes equal contribution; $\alpha - \beta$ denotes alphabetical ordering)

1. A Theory of Entropy Collapse in RLVR
Joe Suk, Yaqi Duan. *In Preparation*.
2. Exploration as Spectral Regularization in Linear Contextual Bandits
Seok-jin Kim*, Joe Suk*, Yaqi Duan. *In Preparation*.
3. [Optimization Dynamics of RLVR: Gradient Gap and Step Size Scaling](#)
Joe Suk, Yaqi Duan. *NeurIPS 2025 Workshop on Foundations of Reasoning in Language Models*. *In Submission*
4. [An Efficient Variant of One-Class SVM with Lifelong Online Learning Guarantees](#)
Joe Suk, Samory Kpotufe. *In submission at Journal of Machine Learning Research (JMLR)*.
5. [Tracking Significant Shifts in Infinite-Armed Bandits](#)
Joe Suk, Jung-hun Kim. *International Conference on Machine Learning (ICML) 2025*.
6. [Adaptive Smooth Nonstationary Bandits](#)
Joe Suk. *SIAM Journal on Mathematics of Data Science (SIMODS)*.
7. [Nonstationary Dueling Bandits with a Weighted Borda Criterion](#)
Joe Suk, Arpit Agarwal. *Transactions on Machine Learning Research (TMLR)* ("Featured Certification").
8. [When Can We Track Significant Preference Shifts in Dueling Bandits?](#)
Joe Suk, Arpit Agarwal. *Advances in Neural Information Processing Systems (NeurIPS) 2023*.
9. [Tracking Most Significant Switches in Nonparametric Contextual Bandits](#)
Joe Suk, Samory Kpotufe. *Advances in Neural Information Processing Systems (NeurIPS) 2023*.
10. [Tracking Most Significant Arm Switches in Bandits](#)
Joe Suk, Samory Kpotufe. *Conference on Learning Theory (COLT) 2022*.
11. [Self-Tuning Bandits over Unknown Covariate-Shifts](#)
Joe Suk, Samory Kpotufe. *International Conference on Algorithmic Learning Theory (ALT) 2021*.
12. [Dihedral Sieving Phenomena](#)
($\alpha - \beta$) Sujit Rao, Joe Suk. *Discrete Mathematics*.
13. [Factorizations of \$k\$ -Nonnegative Matrices](#)
($\alpha - \beta$) Sunita Chepuri, Neeraja Kulkarni, Joe Suk, Ewin Tang. *Journal of Combinatorics*.

AWARDS

- DeepMind student travel grant for COLT 2022.
- William Lowell Putnam Math Competition Top 500.
- Kuga-Sah Memorial Award in Mathematics for outstanding junior, senior math major at Stony Brook University.
- Srivastav, Tucker & Weitzman Scholarship in Applied Mathematics.

ACADEMIC SERVICE AND OUTREACH

- Reviewing/Refereeing (80 papers in total):
 - Journals: JRSS-B, JMLR, TMLR, IEEE Trans. Inf. Theory, Enumerative Combinatorics & Applications.
 - Conferences: NeurIPS ("Top Reviewers" in 2023, 2025), AISTATS, ICML, IJCAI, ICLR, COLT.
- Designed and taught core competency exam [review sessions](#) for Columbia PhD Statistics students in 2021 and 2022.
- Graduate student mentor for [Columbia Summer REU in Mathematical Modeling](#) in 2021 and 2022.
- Teaching Assistant for 20 undergrad/grad courses in statistics and mathematics across Columbia and Stony Brook.

EARLIER RESEARCH EXPERIENCE

Data Science Intern at Institute for Pure and Applied Mathematics (IPAM) *Summer 2018*

- Developed data science pipeline in Python to model microstructure evolution in 3D printing.

Undergraduate Mathematics Honors Thesis advised by Prof. Chris Bishop *2017–2018*

- Developed algorithm to approximate planar trees using harmonic measure and dessins d'enfant.

University of Minnesota Twin Cities Combinatorics NSF REU *Summer 2017*

- Worked on two published research projects in combinatorics and representation theory.

Stony Brook University Geometry/Topology NSF REU *Summer 2016*

- Developed algorithm to count the mapping class group orbits of geodesics on the hyperbolic punctured torus.

TECHNICAL SKILLS

- Programming: Python, Julia, zsh/Bash.
- Other Technical: slurm, git, Linux sysadmin (Artix/Arch Linux and Ubuntu), $\LaTeX$, Claude Code, Codex.

REFERENCES

- Prof. Samory Kpotufe (PhD advisor; Columbia Statistics Department): samory@columbia.edu
- Dr. Yasin Abbasi-Yadkori (former collaborator/mentor; formerly at Google DeepMind, now at Sapien Intelligence): yasin.abbasi@gmail.com